

Sandra Trajkovski

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SUMMARY

Biomedical engineer with hands-on experience in device design, prototyping, and iterative testing across academic research and clinical settings. Proficient in SolidWorks and AutoCAD with a background in microfluidic chip fabrication, fixture and test setup design, and instrument build-and-validate from scratch. Comfortable contributing to cross-functional teams, taking ownership of assigned tasks, and applying curiosity and problem-solving to early-stage development work.

EDUCATION

University of California, Berkeley Aug 2025 – May 2026
Master of Engineering in Bioengineering | Magna Cum Laude

California State University, Long Beach Aug 2021 – May 2025
Bachelor of Science in Biomedical Engineering | Summa Cum Laude | Engineering Honors Program

EXPERIENCE

Machine Learning Engineer – EEG & Biosignal Processing San Francisco, CA
University of California, San Francisco Health | Amorim Lab Aug 2025 – May 2026

- Developed and validated multi-stage engineering pipelines for a large-scale clinical dataset; performed root cause analysis on design and output issues and iterated to resolution.
- Prepared technical documentation and reports summarizing methods, results, and recommendations; presented findings to cross-functional clinical and engineering collaborators.

Student Researcher – Microfluidics & Biological Systems Long Beach, CA
Autonomous Machinery for Engineering Living Matter Lab, CSULB Apr 2024 – May 2025

- Designed and fabricated microfluidic device prototypes from concept through testing; created device layouts, selected materials, and built assemblies using laser cutting and manual fabrication techniques.
- Iterated on designs based on experimental results and performance data; troubleshooted fabrication issues and proposed design changes to improve device function under controlled flow conditions.
- Designed and assembled test fixtures and flow setups to support process validation experiments using structured design-of-experiments (DOE) trials; documented fabrication procedures and test methods for reproducibility across the study.
- Collaborated with PI and lab team on project direction; contributed to design discussions, adapted to evolving experimental requirements, and managed multiple tasks concurrently.

Teacher Assistant – Biomedical Signal Processing Long Beach, CA
California State University, Long Beach Aug 2024 – Dec 2024

- Developed instructional materials and guided 30+ students through hands-on lab work in instrumentation and data analysis; communicated technical concepts clearly to varied backgrounds.

PROJECTS

EEG-Based Neurological Outcome Prediction (MEng Capstone) Spring 2026

- Designed and executed a full clinical engineering pipeline, achieving 93% AUC-ROC in predicting neurological recovery across a dataset of ~1,000 patients; applied iterative development and structured testing to refine outputs, with formal reporting to clinical and engineering stakeholders.

VR-Integrated Low-Cost EEG Brain-Computer Interface Spring 2025

- Contributed to a team building an IRB-approved biosignal acquisition system audited against ISO 13485, ISO 14971, and IEC 62304; handled hardware assembly, debugging, and signal processing pipeline implementation.

Lead Contamination Detection with Spectrophotometry Fall 2024

- Designed and built a functioning spectrophotometer from components - sourced parts, created mechanical layout, assembled the instrument, and validated performance against standards. Full concept-to-validation build.

SKILLS

CAD & Design: SolidWorks, AutoCAD - 3D modeling, assemblies, engineering drawings, GD&T

Fabrication & Prototyping: Microfluidic device fabrication, laser cutting, fixture design, bench assembly, test setup design, instrument build-and-validate

Analysis & Documentation: Statistical analysis, root cause analysis, technical report writing, SOP development, MATLAB, Python, Microsoft Excel, Word, PowerPoint

Professional: Cross-functional collaboration, iterative design, project scheduling, attention to detail, independent and team-based project work